CODE SPORT

Sandya Mannarswamy

In this month's column, we discuss the topic of robotic process automation.

Happy New Year to all our readers! As we start a new year, this month's column also is about a new topic. One of our readers had requested me to cover robotic process automation (RPA). Hence, this month's column provides a brief overview of this interesting and important trend in information technology. We will also examine how natural language processing and machine learning play a role in RPA.

What is robotic process automation?

Robotic process automation (RPA) is defined by the Institute for Robotic Process Automation (IRPA) as 'the application of technology allowing employees in a company to configure computer software or a 'robot' to capture and interpret existing applications for processing a transaction, manipulating data, triggering responses and communicating with other digital systems.' In other words, it involves automating certain tasks or processes that manipulate data. While the definition of RPA as defined by IRPA is very broad and uses the word 'robot', real life RPA has nothing to do with robots. Though the term 'robotic process automation' conjures up visions of an army of robots doing some human tasks, such as moving things or carrying out some other such physical labour, RPA does not have anything to do with robots themselves. It is a form of automation where intelligent software processes take over certain tasks that have typically been performed by human beings.

The readers may now be wondering how this is different from normal software processes involved in desktop automation, which typically do many tasks such as generating bar charts from data, or performing accounting calculations in Excel spreadsheets, typically performed by defining custom macros. Well, the difference comes from the fact that such simple automation scripts fail when there is a decision to be made and when there are a set of complicated steps to follow in order to achieve a task. Typically, desktop automation is limited to defining functions or macros that operate within an application. Let us consider an example from the financial services, namely, the accounts payable process. This is a critical function in all organisations. It includes reconciling the statements submitted by the vendors against the statements of the internal buying departments of an organisation. An organisation can buy goods and services from outside vendors. The purchase process is typically carried out by means of purchase orders. When the goods and services are received, the buying departments create the 'received' reports. The vendors or sellers of these goods and services then submit invoices for payment by the organisation. The reconciliation is carried out by a three-way process between the vendor invoices, received reports and the purchase orders.

While this appears to be a routine task, mismatches between these three data sources can require human intervention and decision making. A typical mismatch could be where the vendor shows the invoice as unpaid whereas the internal department system shows the invoice as paid. The accounts payable process then needs to identify the root cause of the mismatch, and so carries out one of the following three actions. If it finds that the invoice has already been paid, it needs to send the payment details to the vendor. If it has been processed and marked as not paid, it needs to provide the reasons for non-payment to the vendor. Or if the invoice has not been received, it should send a notification to the vendor to send a duplicate